# Zhenzhe Lin

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**Authorized to Work in the U.S. | No Employer Sponsorship Required**

## Research Interests

My research lies at the intersection of **wireless and mobile systems, artificial intelligence, and edge computing**, with a focus on intelligent, efficient, and adaptive networked systems.

- **AI-Driven Wireless and Mobile Systems:** learning-based wireless communication, intelligent network adaptation, and AI-assisted signal processing.
- **Edge AI and Resource-Efficient Intelligence:** efficient machine learning inference, model compression, and latency-aware computing for resource-constrained systems.
- **Battery-Free Communication and Sensing:** backscatter networking, low-power IoT, mmWave communication, and ubiquitous wireless sensing.

## Education

**George Mason University**, Fairfax, VA **2021–Expected Spring 2027**
Ph.D. Candidate in Computer Science
Advisor: Prof. Parth Pathak

**New Jersey Institute of Technology**, Newark, NJ **2018**
M.S. in Electrical Engineering

**Dalian Maritime University**, Dalian, China **2016**
B.E. in Electrical Engineering

## Peer-Reviewed Conference Publications

1. **Zhenzhe Lin**, Yoon Chae, Panneer Selvam Santhalingam, Mingyo Jeong, and Parth Pathak. "Wideband Low-complexity High-speed 5G NR Backscatter." *Proceedings of ACM MobiSys*, 2026.
2. **Zhenzhe Lin**, Yoon Chae, Mingyo Jeong, and Parth Pathak. "$B^3$: Bistatic Backscatter Beamforming for mmWave IoTs." *Proceedings of ACM/IEEE SenSys*, 2026.
3. Yoon Chae, **Zhenzhe Lin**, Kang Min Bae, Song Min Kim, and Parth Pathak. "mmComb: High-speed mmWave Commodity WiFi Backscatter." *Proceedings of USENIX NSDI*, 2024.
4. **Zhenzhe Lin**, Yucheng Xie, Xiaonan Guo, Yanzhi Ren, Yingying Chen, and Chen Wang. "WiEat: Fine-grained Device-free Eating Monitoring Leveraging Wi-Fi Signals." *Proceedings of IEEE ICCCN*, 2020.
5. **Zhenzhe Lin**, Yucheng Xie, Xiaonan Guo, Chen Wang, Yanzhi Ren, and Yingying Chen. "WiFi-enabled Automatic Eating Moment Monitoring Using Smartphones." *Proceedings of EAI HealthyIoT*, 2019. **Best Paper Award**.

## Posters and Research Presentations

- Yoon Chae, **Zhenzhe Lin**, and Parth Pathak. "mmWave WiFi Backscatter Communication." Commonwealth Cyber Initiative (CCI) Symposium, Richmond, VA, 2024.

## Academic Research Experience

**George Mason University — Graduate Research Assistant** Jan. 2021–Present

**Learning-Driven Wideband 5G NR Backscatter Communication** *MobiSys 2026, First Author*

- **AI Model Design:** Independently designed and implemented a multi-task transformer architecture that jointly performs wireless channel estimation and backscatter signal decoding, addressing channel uncertainty and low-SINR reception in wideband 5G NR networks.

- **Efficient AI Inference:** Developed and evaluated model pruning and inference optimization techniques to reduce computational overhead and inference latency while maintaining reliable decoding performance.
- **End-to-End Systems Research:** Integrated the learning-based receiver into a practical 5G NR backscatter system and conducted experimental evaluations of communication reliability, computational efficiency, and real-world feasibility.

**Tag-Assisted mmWave Beamforming and Channel Sensing** *SenSys 2026, First Author*

- Proposed a battery-free, tag-assisted wireless architecture for concurrent multi-user channel sensing and beamforming, reducing the need for active client-side beam training.
- Developed backscatter channel isolation, angle-of-departure estimation, and beam-weight optimization algorithms to enable efficient mmWave beamforming.
- Built and evaluated a prototype using custom backscatter tags and commodity wireless devices under multipath, mobility, and blockage conditions.

**High-Speed Backscatter over Commodity mmWave Wi-Fi** *NSDI 2024, Second Author*

- Contributed to a high-speed battery-free communication architecture integrating backscatter devices with commodity IEEE 802.11ad Wi-Fi infrastructure.
- Developed spatial interference management techniques using beamforming and nulling to improve reception of weak backscatter signals.
- Implemented and evaluated a mmWave backscatter prototype, demonstrating data rates up to 55 Mbps and communication over meter-scale distances.

**Rutgers University, WINLAB — Research Assistant** Sep. 2019–Dec. 2020

**Wi-Fi and RFID Sensing Systems**

- Studied environment-invariant detection of concealed objects using commodity Wi-Fi channel state information and adversarial representation learning.
- Investigated RFID-array sensing for heartbeat and heart-rate variability estimation, including signal modeling, denoising, and sensor fusion.
- Developed mmWave-based 3D skeletal posture reconstruction methods using neural networks and domain-invariant feature learning.

**Rutgers University, WINLAB — Research Intern** Aug. 2018–Aug. 2019

- Developed device-free eating activity monitoring using Wi-Fi channel state information, motion reconstruction, and machine-learning-based activity recognition; resulting work appeared at ICCCN 2020 and HealthyIoT 2019.

## Teaching Experience

- **Teaching Assistant**, CS 531: Fundamentals of Systems Programming, George Mason University Fall 2022
- **Teaching Assistant**, CS 112: Introduction to Computer Programming, George Mason University Spring 2021
- **Teaching Assistant**, Semiconductor Device I, Rutgers University Fall 2020
- **Teaching Assistant**, Physical Electronics, Rutgers University Fall 2020

## Student Mentoring

- **Mentor**, WINLAB Summer Internship Program, Rutgers University Summer 2020
- **Mentor**, ECE Senior Design Capstone Projects, Rutgers University Spring 2020

## Honors, Awards, and Grants

- **Doctoral Research Scholarship – Tier 1**, George Mason University ($25,000) 2026–2027
- **Dissertation Completion Grant**, George Mason University ($12,500) Spring 2026
- **Best Paper Award**, EAI HealthyIoT 2019

## Technical Expertise

**Wireless and Experimental Systems:** 5G NR, mmWave Wi-Fi, RF sensing, backscatter systems, software-defined radio, FPGA prototyping.
**Programming and Machine Learning:** Python, MATLAB, C++, PyTorch, TensorFlow, Verilog, VHDL.